

1. Problem Understanding

Description of the Problem

The objective of this project is to build classification models to predict diabetes risk using patient health data such as age, BMI, blood pressure, exercise habits, and smoking status.

Two approaches are used:

- **A Machine Learning model** (Logistic Regression)
- **A Deep Learning model** (Artificial Neural Network)

The goal is to compare their performance and understand which model is more suitable for structured healthcare data.

Importance of Predicting Diabetes Risk

Predicting diabetes early is important because:

- Enables early intervention and lifestyle changes
- Reduces long-term healthcare costs
- Improves patient outcomes
- Helps in public health planning
- Supports data-driven medical decisions

2. Data Exploration Insights

Dataset Overview

- **Total records:** 6000 rows × 7 columns
- **No missing values** in dataset
- **Target variable:** diabetes (0 = No, 1 = Yes)

Key Observations

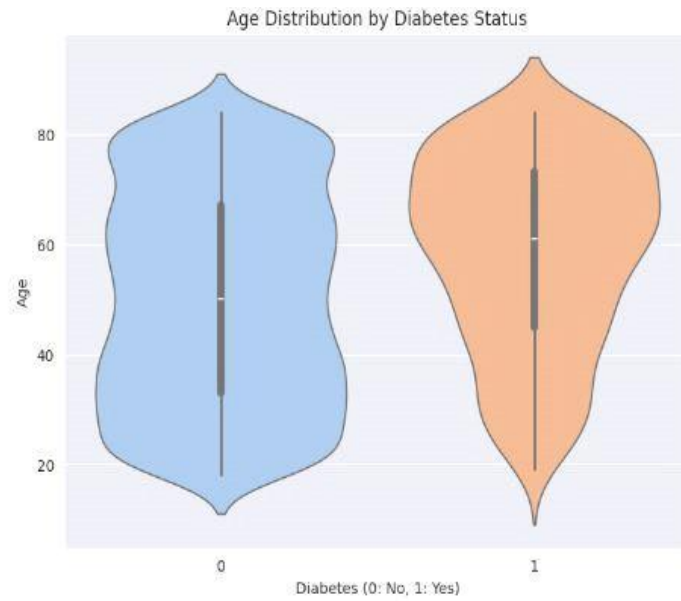
From the analysis:

Class Imbalance

- **Non-diabetic:** 5458
- **Diabetic:** 542 → Only ~9% positive cases (important insight)

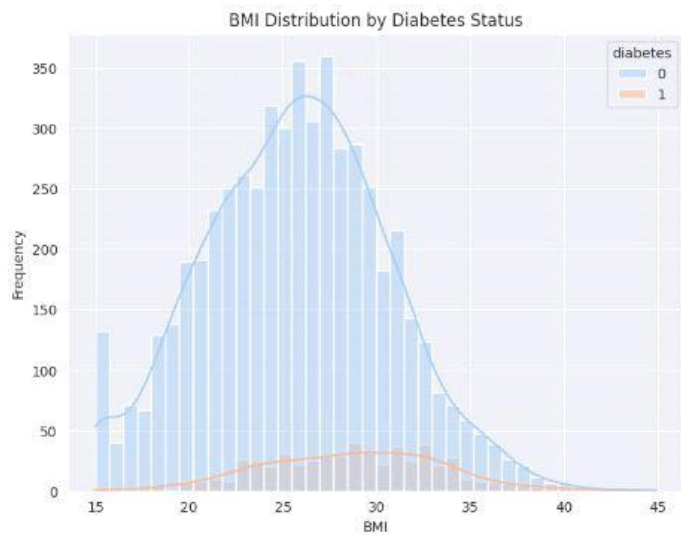
Age

- **Range:** 18–84
- Higher diabetes cases in 60–80 age group



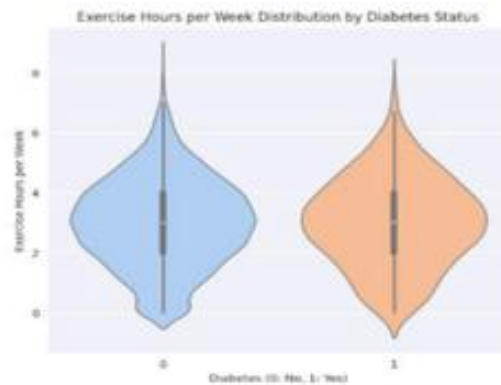
BMI

- **Average:** ~26
- Risk increases significantly when BMI > 30



Exercise

- Lower activity → higher diabetes risk



Blood Pressure

- Higher BP correlates with diabetes

Feature Importance (from findings)

- **BMI** → Strongest predictor
- **Blood Pressure** → Moderate impact
- **Age** → Significant demographic factor

3. Model Implementation

3.1 Machine Learning Model — Logistic Regression

Steps Performed:

- Feature selection (X) and target (y)
- Train-test split
- Feature scaling
- Model training
- Prediction

Model Explanation

Logistic Regression is a linear model that:

- Predicts probability using weighted features
- Is easy to interpret
- Works well for structured/tabular data

3.2 Neural Network Model (ANN)

Architecture Used

- **Input Layer** → 5 features
- **Hidden Layer 1** → 16 neurons (ReLU)
- **Hidden Layer 2** → 8 neurons (ReLU)

- **Output Layer** → 1 neuron (Sigmoid)

Compilation

- **Optimizer:** Adam
- **Loss:** Binary Crossentropy

Training

- Model trained for multiple epochs
- Evaluated on test dataset

4. Model Evaluation

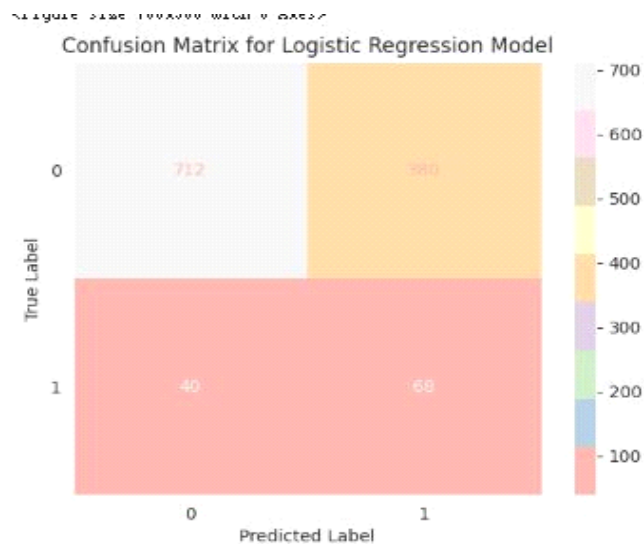
4.1 Logistic Regression Performance

- **Accuracy:** 65%

Interpretation:

- Moderate performance
- More balanced predictions
- Better interpretability

Confusion Matrix — Logistic Regression:



4.2 Neural Network Performance

- **Accuracy:** 90.5%

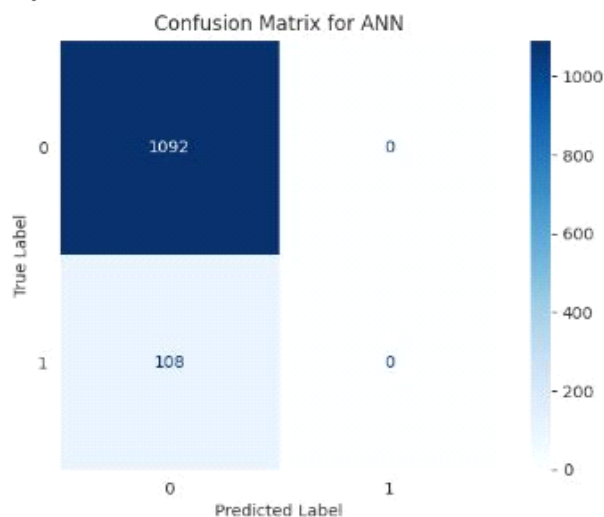
Classification Report (Important Insight)

High accuracy but:

- **Recall (Diabetes = 1):** 0.04
- Model fails to detect diabetic cases

Confusion Matrix Insight

- Model predicts mostly non-diabetic
- Poor detection of actual diabetic patients



⚠ Critical Observation: Even though ANN has higher accuracy, it is misleading due to class imbalance.

5. Comparative Analysis

Accuracy Comparison

Model	Accuracy
Logistic Regression	65%
Neural Network (ANN)	90.5%

Why Neural Network Shows Higher Accuracy

- Dataset is imbalanced
- Model learns majority class (non-diabetic)
- Ignores minority class (diabetic)

Strengths of Logistic Regression

- Interpretable
- Simple and fast
- Works well on tabular data
- Better for healthcare (explainable)

Strengths of Neural Network

- Captures complex patterns
- Higher accuracy (in balanced datasets)
- Powerful for large data

Limitations

Logistic Regression

- Cannot capture complex relationships
- Lower accuracy

Neural Network

- Black-box model
- Needs more data
- Sensitive to imbalance
- Poor interpretability

6. Recommendations

- Use Logistic Regression for healthcare systems (due to interpretability)

Improve Neural Network using:

- Class balancing (SMOTE / class weights)
- Better evaluation metrics (Recall, F1)
- Focus on Recall (Diabetes detection) instead of accuracy

7. Conclusion

This study shows that:

- Higher accuracy does not always mean better model
- Neural networks can be misleading with imbalanced data
- Logistic Regression is more reliable for healthcare prediction tasks